

Multipoint Newton–Okounkov Bodies and Jet Separation

Doctoral Thesis Outline

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Structure of the Talk

The story moves from definitions, to criteria, and then to the double-cover tests.

- ① Notation and definitions
- ② Jet-separation criteria
- ③ Examples and calculations

Notation

The main symbols are the variety, divisors, sections, and local ideals.

X smooth projective variety, $\dim X = d$, over $\mathbb{C}$

D divisor on X

K_X canonical divisor

$H^0(X, \mathcal{O}_X(D))$ vector space of global sections of $\mathcal{O}_X(D)$

$\mathfrak{m}_x$ maximal ideal of $\mathcal{O}_{X,x}$ at a smooth point x

Definitions and Motivation

Jet Separation and Jet Ampleness

Jet ampleness can be viewed as a uniform collection of jet-separation conditions with prescribed total weight.

Definition (Jet separation and jet ampleness)

Let L be a line bundle on X . For distinct points $x_1, \dots, x_r$ and positive weights $a_1, \dots, a_r$, consider

$$H^0(X, L) \longrightarrow H^0(X, L \otimes \mathcal{O}_X / (\mathfrak{m}_{x_1}^{a_1} \cdots \mathfrak{m}_{x_r}^{a_r})).$$

- Surjectivity is jet separation for the fixed weighted configuration $(x_1, \dots, x_r; a_1, \dots, a_r)$.
- When $r = 1$, this is the usual jet separation at one point.
- For a fixed weight distribution $\sum_i a_i = k + 1$, requiring this for all choices of distinct points gives one part of k -jet ampleness.
- L is k -jet ample if this holds for every such weight distribution.
- 0-jet ampleness = global generation; 1-jet ampleness = very ampleness.

Newton–Okounkov Body

A Newton–Okounkov body records asymptotic sections through valuation vectors.

Definition (Ordinary Newton–Okounkov body)

Let D be a big divisor on a d -dimensional variety X . An admissible flag centered at x is

$$X = Y_0 \supset Y_1 \supset \cdots \supset Y_d = \{x\},$$

where $\operatorname{codim} Y_i = i$ and each Y_i is smooth at x . It defines

$$\nu_{Y_\bullet}(s) = (\nu_1(s), \dots, \nu_d(s)) \in \mathbb{Z}^d, \quad 0 \neq s \in H^0(X, \mathcal{O}_X(mD)).$$

The Newton–Okounkov body is

$$\Delta_{Y_\bullet}(D) = \operatorname{Cone}\{(\nu_{Y_\bullet}(s), m) : 0 \neq s \in H^0(X, \mathcal{O}_X(mD)), m \geq 1\} \cap (\mathbb{R}^d \times \{1\}).$$

Variants of Newton–Okounkov Bodies

The variants change the flags, and multipoint versions add a competition rule.

Definition (Infinitesimal and multipoint variants)

Starting from the ordinary construction:

- **Infinitesimal.** Replace the flag by one on the blow-up $\pi : \tilde{X} \rightarrow X$:

$$\tilde{X} = \tilde{Y}_0 \supset \tilde{Y}_1 = E \supset \cdots \supset \tilde{Y}_d = \{q\}.$$

Then take $\Delta_{\tilde{Y}_\bullet}(\pi^*D)$.

- **Multipoint.** Use several flags/valuations ν_j centered at $p_j \in Z = \{p_1, \dots, p_N\}$, and keep a section only where its valuation is strictly smallest:

$$V_{r,j} = \{0 \neq s \in H^0(X, rL) : \nu_j(s) < \nu_i(s) \text{ for all } i \neq j\}.$$

This produces one body $\Delta_j(L)$ for each point. The flags may be **normal/infinitesimal** flags on $X/\tilde{X} = \text{Bl}_Z(X)$.

Inverted Standard Simplex

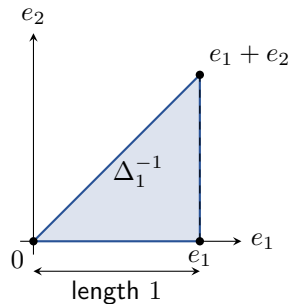
The inverted simplex is the small convex shape whose containment gives the required local positivity size.

Definition (Inverted standard simplex)

For $\xi \geq 0$,

$$\Delta_{\xi}^{-1} = \text{Conv}\{0, \xi e_1, \xi(e_1 + e_2), \dots, \xi(e_1 + \dots + e_d)\} \subset \mathbb{R}^d.$$

Containment of Δ_{ξ}^{-1} in an infinitesimal Newton–Okounkov body is the convex-geometric form of local positivity of size ξ .



Multipoint Example: $\mathbb{P}^2$ with $\mathcal{O}(3)$

This toric example gives a concrete picture of that competition.

Toric visualization inside the multipoint definition

Let $X = \mathbb{P}^2$ and $L = \mathcal{O}(3)$. The associated polytope is

$$\Delta = \text{Conv}\{(0, 0), (3, 0), (0, 3)\} \subset \mathbb{R}^2.$$

The three selected torus-fixed points correspond to the three vertices. For the flags centered at these points with lexicographic monomial order, the local coordinate maps are

$$\varphi_1(x, y) = (x + y, x), \quad \varphi_2(x, y) = (-x, y), \quad \varphi_3(x, y) = (-y, -x - y).$$

Trusiani's toric subdivision theorem identifies the valid parts Δ_j° with a subdivision of Δ .

Figure 3: Toric Visualization on One Page

Read this page from left to right: original polytope, local-chart pictures, normalized infinitesimal bodies, and the final comparison under two partial orders.

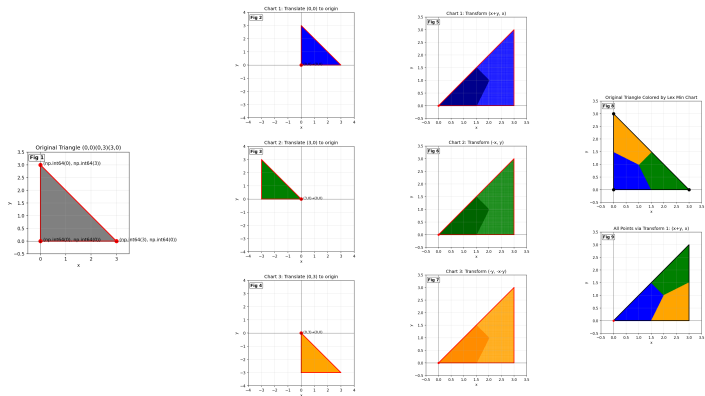


Figure: Left: the original polytope Δ . Second column: the same global sections in the three local charts realizing the corresponding valuations. Third column: the simultaneously computed infinitesimal bodies after normalization; the dark part is the region assigned to that flag by the multipoint competition rule. Fourth column: the resulting pictures for two different partial-order types.

Jet-Separation Criteria

Lemmas Used in the Criteria

The criteria depend on three mechanisms: nonvanishing multiplication, comparison with one-point bodies, and slicing.

Lemma 2.1

If A has a section nonvanishing on $Z = \{p_1, \dots, p_N\}$, multiplication by powers of this section preserves all infinitesimal valuation vectors over the points of Z .

► Proof

Lemma 2.3

For every tuple of infinitesimal flags over Z and every j ,

$$\Delta_{\tilde{Y}_{\bullet}, j}(\pi^* D) \subseteq \Delta_{\tilde{Y}_{\bullet}^{(j)}}(\pi^* D).$$

► Proof

Lemma 2.4

If the j th multipoint infinitesimal body of $\pi^* D$ contains points approaching $(a, 0, \dots, 0)$ from $x_1 > a$, then the origin belongs to the corresponding one-point infinitesimal body of $\pi^* D - aE$.

► Proof

Multipoint Adjoint Jet-Separation Criterion

The multipoint criterion asks for the inverted simplex inside every multipoint infinitesimal body.

Theorem 2.5

Let X be a d -dimensional smooth projective variety, let D be a big Cartier divisor, and let $Z = \{p_1, \dots, p_N\}$ be a set of distinct smooth points. Let $\pi : \tilde{X} = \text{Bl}_Z(X) \rightarrow X$ be the blow-up. If there are $k \in \mathbb{N}$ and $\varepsilon > 0$ such that

$$\Delta_{d+k+\varepsilon}^{-1} \subseteq \Delta_{\tilde{Y}_{\bullet,j}}(\pi^*D)$$

for every $j = 1, \dots, N$ and every tuple of infinitesimal flags over Z , then $K_X + D$ separates k -jets simultaneously at $p_1, \dots, p_N$.

► Proof

Canonical-Free Multipoint Criterion

The canonical-free version applies the containment to a multiple of the divisor, then adds one global-generation correction.

Theorem 2.6

Let D be ample and let $Z = \{p_1, \dots, p_N\}$. Choose $m(D)$ with $m(D)D - K_X$ globally generated. If, for some $m > 0$ and some $\varepsilon > 0$,

$$\Delta_{d+k+\varepsilon}^{-1} \subseteq \Delta_{\tilde{Y}_{\bullet,j}}(\pi^*(mD))$$

for every j and every tuple of infinitesimal flags over Z , then $(m + m(D))D$ separates k -jets simultaneously at $p_1, \dots, p_N$.

► Proof

Single-point case

When $Z = \{x\}$, this gives Theorem 2.2: the same containment for every infinitesimal flag over x implies that $(m + m(D))D$ separates k -jets at x .

► Proof

Examples and Calculations

Double Covers of Products of Elliptic Curves

The examples now test the criteria in a surface family where intersection numbers can be computed explicitly.

The surface and the ample divisors

Let $S = E \times E$, where E is an elliptic curve. Let F_1, F_2 be the two fibre classes and let Δ be the diagonal. Their intersection numbers are

$$F_1^2 = F_2^2 = \Delta^2 = 0, \quad F_1 \cdot F_2 = F_1 \cdot \Delta = F_2 \cdot \Delta = 1.$$

Define

$$A_n = nF_1 + (n^2 - n + 1)F_2 - (n - 1)\Delta.$$

If $H = F_1 + F_2$, then

$$A_n^2 = 2, \quad A_n \cdot H = n^2 - 2n + 3 > 0.$$

The Double Cover and D_n

The double cover transfers the ample divisors from the abelian surface to a surface with nontrivial canonical class.

Double cover

Let $R = F_1 + F_2$, choose a smooth divisor in $|2R|$, and let

$$f : Y \rightarrow S$$

be the double cover branched along it. Define

$$D_n = f^* A_n.$$

Since f is finite and A_n is ample, D_n is ample. The canonical divisor is computed by

$$K_Y = f^*(K_S + R) = f^* R,$$

because $K_S = 0$.

Figure: Schematic for the Double-Cover Example

This diagram separates the downstairs blow-up from the two upstairs branches over a non-branch point.

$$\begin{array}{ccccc}
 \pi_x^* A_n & E_x & \longleftarrow & \tilde{E}_{y_1} \sqcup \tilde{E}_{y_2} & \tilde{f}^* \pi_x^* A_n \\
 & & & \cap & \parallel \\
 & & \tilde{f} & & \pi_{1,2}^* D_n \\
 \text{Bl}_x S & \longleftarrow & & \text{Bl}_{y_1, y_2} Y & \\
 & 2-1 & & & \\
 & \downarrow \pi_x & & \downarrow \pi_2 & \\
 & & E_1 \subseteq & \text{Bl}_{y_1} Y & \pi_1^* D_n \\
 & & & \downarrow \pi_1 & \\
 A_n & & & & D_n = f^* A_n \\
 \downarrow & & & & \\
 x \in S = E \times E & \longleftarrow & f & Y & \{y_1, y_2\} = f^{-1}(x) \\
 & 2-1 & & & \\
 \cup & & & & \\
 x \notin B & \in & |2(F_1 + F_2)| & &
 \end{array}$$

Figure 1: Schematic for the double-cover example.

Seshadri Constants at Non-Branch Points

At non-branch points the map is étale, so downstairs local data can be compared directly upstairs.

Proposition 4.1

For every point $x \in S$,

$$\varepsilon(A_n; x) = 1.$$

If $x \in S \setminus B$ and $f^{-1}(x) = \{y_1, y_2\}$, then

$$\varepsilon(D_n; y_i) = 1, \quad i = 1, 2.$$

► Proof

Non-Branch Body Inclusion

With the local comparison in place, the valuation bodies can be compared through the pulled-back linear series.

Proposition 4.2

For a non-branch point $x \in S \setminus B$ and $y_i \in f^{-1}(x)$,

$$\Delta_{S^\bullet_x}(\pi_x^* A_n) = \Delta_{Y^\bullet_{y_i}}(W_\bullet) \subseteq \Delta_{Y^\bullet_{y_i}}(\pi_{y_i}^* D_n).$$

Consequently, for every infinitesimal flag over y_i ,

$$\Delta_1^{-1} \subseteq \Delta_{Y^\bullet_{y_i}}(\pi_{y_i}^* D_n).$$

► Proof

Local Jet-Separation Bound

The local bound applies the canonical-free criterion at a fixed non-branch point.

The key numerical input is

$$N'_n = A_n \cdot R = n^2 - 2n + 3, \quad m_{gg}(D_n) \leq 2N'_n.$$

Local bound

For $x \in S \setminus B$ and $y \in f^{-1}(x)$,

$$(2N'_n + 4)D_n = (2n^2 - 4n + 10)D_n$$

separates 1-jets at y .

► Proof

Newton–Okounkov Body for a Concrete Flag

For this flag, the body is controlled by one endpoint and one linear upper boundary.

Choose

$$Z_1 \in |2F_1 + 2F_2 + \Delta|, \quad Y \supset f^*Z_1 \supset \{y\}.$$

The key quantities are

$$N_n = A_n \cdot Z_1 = 3n^2 - 4n + 7, \quad T(n) = \frac{N_n}{16} \left(1 - \sqrt{1 - \frac{32}{N_n^2}} \right).$$

Computed body

$$\Delta_{Y_\bullet}(D_n) = \{(t, u) : 0 \leq t \leq T(n), 0 \leq u \leq 2(N_n - 16t)\}.$$

► Proof

Figure: Shape of $\Delta_{Y_\bullet}(D_n)$

The narrow first-coordinate width is the feature that matters when compared with global bounds.

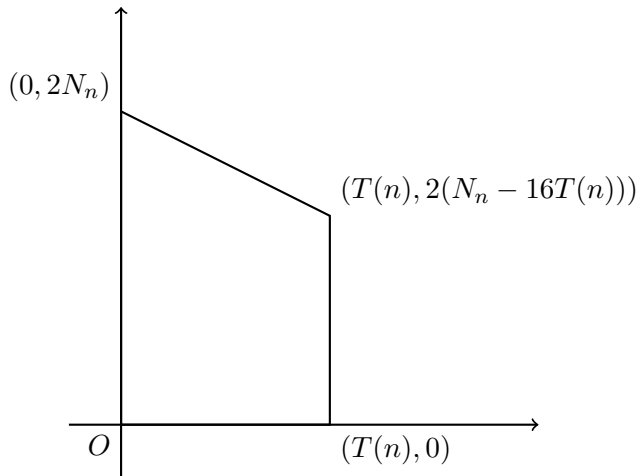


Figure 2: Shape of $\Delta_{Y_\bullet}(D_n)$ for the flag $Y \supset f^*Z_1 \supset \{y\}$.

Global Jet-Ampleness Bound

The global statement is stronger because it has to handle all point configurations.

The key integer is

$$N''_n = \left\lceil \frac{N'_n + \sqrt{(N'_n)^2 - 4}}{2} \right\rceil = N'_n \quad (n \geq 2).$$

Global jet-ampleness bound

$$(m + N''_n)D_n$$

is m -jet ample for every $m \geq 1$.

► Proof

Ramification and Local Valuations

At a branch point the étale comparison breaks down, and the valuation records the ramification weight.

Proposition 4.4

Let $x \in B$ be a smooth branch point and let y be the unique point of Y over x . Choose formal parameters (u, v) at x such that $B = (v = 0)$. Then, after completion,

$$\widehat{\mathcal{O}}_{Y,y} \simeq \mathbb{C}[[u, w]], \quad v = w^2.$$

For a local function

$$g(u, v) = \sum_{i,j} a_{ij} u^i v^j,$$

one has

$$f^*g(u, w) = \sum_{i,j} a_{ij} u^i w^{2j}, \quad \text{ord}_y(f^*g) = \min\{i + 2j : a_{ij} \neq 0\}.$$

Thus the branch case corresponds to the $(1, 2)$ -weighted valuation in the parameters (u, v) .

► Proof

Intrinsic Positivity at Branch Points

At branch points, intrinsic positivity on Y is still controlled by the Seshadri constant.

Corollary 4.5

For $y \in Y$ over a smooth branch point and any infinitesimal flag over y ,

$$\xi(D_n; y) = \varepsilon(D_n; y).$$

The negative part of the Zariski decomposition of

$$\pi_y^* D_n - tE_y$$

is zero for every $0 \leq t < \varepsilon(D_n; y)$.

► Proof

Thank you

Appendix: Detailed Proofs

Proof.

Multiplication by a^r gives

$$M_r : H^0(X, rD) \rightarrow H^0(X, r(D + A)), \quad s \mapsto s \otimes a^r.$$

Since $a(p_i) \neq 0$ for all i , its pullback has valuation $\mathbf{0}$ along every infinitesimal flag over Z . Hence

$$\nu_i(M_r(s)) = \nu_i(s) \quad \forall i.$$

For multipoint bodies, if $s \in V_{r,j}(D)$, then

$$\nu_j(s) <_{\text{lex}} \nu_i(s) \quad (i \neq j).$$

The equalities above preserve this winning condition, so $M_r(s) \in V_{r,j}(D + A)$. Taking closed normalized convex hulls gives the one-point and multipoint inclusions. □

Appendix: Lemma 2.3

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Proof.

For every $r \geq 1$ and every j , the winning subspace is

$$V_{r,j}(D) = \{s \in H^0(X, rD) : \nu_j(s) <_{\text{lex}} \nu_i(s) \ \forall i \neq j\}.$$

Thus $V_{r,j}(D) \subseteq H^0(X, rD)$. The multipoint valuation semigroup is a subsemigroup of the ordinary one-point valuation semigroup at p_j . Therefore

$$\Delta_{\tilde{Y}_{\bullet},j}(\pi^* D) \subseteq \Delta_{\tilde{Y}_{\bullet}(j)}(\pi^* D).$$



Proof.

Set $B_a = \pi^*D - a\mathbb{E}$. The multipoint slice formula says that subtracting $a\mathbb{E}$ cuts the part with first coordinate at least a and translates it by $-ae_1$.

The assumed points approaching $(a, 0, \dots, 0)$ from $x_1 > a$ give

$$\mathbf{0} \in \Delta_{\tilde{Y}_{\bullet}, j}(B_a).$$

By Lemma 2.3,

$$\Delta_{\tilde{Y}_{\bullet}, j}(B_a) \subseteq \Delta_{\tilde{Y}_{\bullet}^{(j)}}(B_a),$$

so $\mathbf{0} \in \Delta_{\tilde{Y}_{\bullet}^{(j)}}(B_a)$ for every infinitesimal flag over E_j . □

Proof.

This is the case $N = 1$ of Theorem 2.6. When $Z = \{x\}$, the multipoint condition is the ordinary infinitesimal condition at x . The conclusion of Theorem 2.6 becomes

$(m + m(D))D$ separates k -jets at x .



Proof, first part.

It is enough to prove the surjectivity of

$$H^0(X, \mathcal{O}_X(K_X + D)) \rightarrow \bigoplus_{j=1}^N H^0\left(X, \mathcal{O}_X(K_X + D) \otimes \mathcal{O}_{X, p_j} / \mathfrak{m}_{p_j}^{k+1}\right).$$

Let $\mathbb{E} = E_1 + \cdots + E_N$ on $\overline{X} = \text{Bl}_Z(X)$. Then

$$\pi_* \mathcal{O}_{\overline{X}}(-(k+1)\mathbb{E}) = \bigcap_{j=1}^N \mathfrak{m}_{p_j}^{k+1}.$$

Set

$$B = \pi^* D - (d+k)\mathbb{E}.$$

Since $K_{\overline{X}} = \pi^* K_X + (d-1)\mathbb{E}$, one has

$$K_{\overline{X}} + B = \pi^*(K_X + D) - (k+1)\mathbb{E}.$$



Proof, second part.

For each j , the simplex containment gives points in the j th winning body approaching

$$(d + k, 0, \dots, 0)$$

from the side $x_1 > d + k$. Lemma 2.4 with $a = d + k$ gives

$$\mathbf{0} \in \Delta_{\tilde{Y}_{\bullet}^{(j)}}(B)$$

for every infinitesimal flag over E_j . The restricted-base-locus criterion gives

$$\mathbf{B}_{-}(B) \cap E_j = \emptyset \quad \forall j, \quad \mathbf{B}_{-}(B) \cap \mathbb{E} = \emptyset.$$

Thus the asymptotic multiplier ideal is trivial along $\mathbb{E}$:

$$\mathcal{J}(\overline{X}, \|B\|)|_{\mathbb{E}} = \mathcal{O}_{\mathbb{E}}.$$



Proof, final part.

Nadel vanishing gives

$$H^1(\overline{X}, \mathcal{O}_{\overline{X}}(K_{\overline{X}} + B) \otimes \mathcal{J}(\overline{X}, \|B\|)) = 0.$$

Since the multiplier ideal restricts trivially along $\mathbb{E}$, global sections restrict surjectively to the $(k+1)$ -st infinitesimal neighbourhood of $\mathbb{E}$. Under

$$\mathcal{O}_{(k+1)E_j} \simeq \mathcal{O}_{X,p_j} / \mathfrak{m}_{p_j}^{k+1},$$

this is the desired simultaneous k -jet separation of $K_X + D$ at $p_1, \dots, p_N$. □

Proof, first part.

Set

$$A = m(D)D - K_X.$$

Since A is globally generated and Z is finite, one can choose

$$a \in H^0(X, A), \quad a(p_j) \neq 0 \quad \forall j.$$

For every $r \geq 1$, multiplication by a^r sends

$$s \in H^0(X, rmD) \longmapsto s \otimes a^r \in H^0(X, r(mD + A)).$$

Because $\nu_i(\pi^*a) = 0$ for all i , Lemma 2.1 gives

$$\Delta_{\tilde{Y}_{\bullet}, j}(\pi^*(mD)) \subseteq \Delta_{\tilde{Y}_{\bullet}, j}(\pi^*(mD + A)).$$



Proof, second part.

The same inverted simplex is contained in the body of $mD + A$ at every marked point. Since $mD + A$ is big, Theorem 2.5 applied to $mD + A$ shows that

$$K_X + (mD + A)$$

separates k -jets simultaneously at Z . Finally,

$$K_X + (mD + A) = K_X + mD + m(D)D - K_X = (m + m(D))D.$$



Appendix: Proposition 4.1

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Proof.

Let $F_{2,x} \equiv F_2$ be the translate through x . Since

$$\text{mult}_x(F_{2,x}) = 1, \quad A_n \cdot F_2 = n - (n - 1) = 1,$$

one has $\varepsilon(A_n; x) \leq 1$. The standard lower bound for ample integral line bundles on abelian varieties gives the reverse inequality, so $\varepsilon(A_n; x) = 1$.

If $x \notin B$, then $f : (Y, y_i) \rightarrow (S, x)$ is etale. Multiplicities are preserved, so $\varepsilon(D_n; y_i) \geq 1$. The lift through y_i of $F_{2,x}$ has multiplicity 1 and D_n -degree 1, so $\varepsilon(D_n; y_i) \leq 1$. □

Proof.

Etaleness at y_i identifies the completed local rings, the blow-ups, and the corresponding infinitesimal valuation vectors. Hence the pullback series

$$W_m = f^* H^0(S, mA_n)$$

has the same valuation semigroup as the downstairs complete series:

$$\Delta_{S^\bullet_x}(\pi_x^* A_n) = \Delta_{Y^\bullet_{y_i}}(W_\bullet).$$

The inclusion $W_m \subseteq H^0(Y, mD_n)$ gives the body containment. By Proposition 4.1, $\varepsilon(D_n; y_i) = 1$; on smooth surfaces, this gives Δ_1^{-1} in every infinitesimal body over y_i . □

Proof.

Put $L_q = qA_n - R$ and $N'_n = A_n \cdot R = n^2 - 2n + 3$. Then

$$L_q \cdot A_n = 2q - N'_n, \quad L_q^2 = 2q^2 - 2N'_n q + 2.$$

The positive-cone criterion gives the ample threshold

$$q > \frac{N'_n + \sqrt{(N'_n)^2 - 4}}{2}, \quad \alpha_n = N'_n \quad (n \geq 2).$$

Since $2N'_n A_n - R$ is a tensor product of two ample line bundles, $m_{\text{gg}}(D_n) \leq 2N'_n$. Proposition 4.2 gives $\Delta_1^{-1} \subseteq \Delta(\pi_y^* D_n)$; by homogeneity, $4D_n$ contains an inverted simplex of size $4 > 3$. The canonical-free criterion with $k = 1$ gives

$$(2N'_n + 4)D_n = (2n^2 - 4n + 10)D_n.$$



Proof.

Choose $Z_1 \in |2F_1 + 2F_2 + \Delta|$ general with f^*Z_1 irreducible. The intersections are

$$Z_1^2 = 16, \quad N_n = A_n \cdot Z_1 = 3n^2 - 4n + 7.$$

The pseudo-effective endpoint comes from

$$(A_n - tZ_1)^2 = 16t^2 - 2N_nt + 2,$$

so

$$T(n) = \frac{N_n}{16} \left(1 - \sqrt{1 - \frac{32}{N_n^2}} \right).$$

For $0 \leq t \leq T(n)$, the Zariski decomposition is trivial. The surface formula gives upper boundary $2(N_n - 16t)$, hence

$$\Delta_{Y_\bullet}(D_n) = \{(t, u) : 0 \leq t \leq T(n), 0 \leq u \leq 2(N_n - 16t)\}.$$



Proof.

The divisor $\ell A_n - R$ is ample once

$$(\ell A_n - R) \cdot A_n = 2\ell - N'_n > 0, \quad (\ell A_n - R)^2 = 2\ell^2 - 2N'_n\ell + 2 > 0.$$

Equivalently,

$$\ell > \frac{N'_n + \sqrt{(N'_n)^2 - 4}}{2}.$$

For $n \geq 2$,

$$N''_n = \left\lceil \frac{N'_n + \sqrt{(N'_n)^2 - 4}}{2} \right\rceil = N'_n.$$

The Bauer–Szemberg cyclic-cover criterion then gives that $(m + N''_n)D_n$ is m -jet ample for every $m \geq 1$. □

Proof.

Choose parameters (u, v) at the smooth branch point with $B = (v = 0)$. A smooth double cover ramified along a smooth divisor has completed local form

$$\widehat{\mathcal{O}}_{S,x} \simeq \mathbb{C}[[u, v]], \quad \widehat{\mathcal{O}}_{Y,y} \simeq \mathbb{C}[[u, w]], \quad v = w^2.$$

For $g(u, v) = \sum_{i,j} a_{ij} u^i v^j$, pullback gives

$$f^*g(u, w) = \sum_{i,j} a_{ij} u^i w^{2j}.$$

Thus

$$\text{ord}_y(f^*g) = \min\{i + 2j : a_{ij} \neq 0\},$$

which is the $(1, 2)$ -weighted order in the downstairs parameters. □

Appendix: Corollary 4.5

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Proof.

For ample divisors on smooth surfaces, infinitesimal Newton–Okounkov body theory gives equality between the largest inverted simplex constant and the Seshadri constant:

$$\xi(D_n; y) = \varepsilon(D_n; y).$$

If $0 \leq t < \varepsilon(D_n; y)$, then by definition of the Seshadri constant

$$L_t^Y = \pi_y^* D_n - tE_y$$

is nef. A nef divisor has Zariski decomposition with zero negative part, hence $N_t = 0$ for every such t . □